A Statistical Physics Approach to Market Dynamics: Phase Transitions in Local and Digital Vendor Dominance

Ninaad Adhvaryu
Independent Researcher
Delhi, India
Amaira Goyal
Bombay Scottish School, Mahim
Mumbai, India

A physics-grounded model of consumer choice between digital platforms and local vendors is developed, utilizing a structured utility function inspired by statistical mechanics. Customer selection is interpreted as an energy-minimization process governed by an effective potential. A dual-engine simulation framework is presented, comparing a stochastic agent-based model (ABM) with a self-consistent mean-field approximation. Global control parameters—including decision noise (temperature T), information bandwidth (visibility $\gamma \in [0, 1]$), and two social coupling parameters (bandwagon strength ϕ and Ising-type coupling strength J)—are varied systematically to map macroscopic market phases. Tipping points and regimes of dominance are identified through order parameters including local market share, magnetization, and Shannon entropy. Phase boundaries are located via peaks in the susceptibility $\chi_\gamma = \partial L^*/\partial\gamma$ and $\chi_T = \partial L^*/\partial T$. The ABM and mean-field engine agree in bulk phases but differ qualitatively near phase boundaries: γ acts as a genuine dynamical control in the ABM through its accumulating-knowledge mechanism, whereas the mean-field static-mask approximation largely suppresses χ_γ. The implications for market stability and information-driven transitions between Local Dominance, Digital Dominance, and Mixed Equilibrium are discussed.

Keywords: agent-based modeling; statistical mechanics; phase transitions; digital marketplaces; mean-field theory; discrete choice; market concentration

1. Introduction

1.1 Statistical Physics as a Language for Socioeconomic Systems

A central insight of modern statistical mechanics is that the macroscopic properties of many-body systems are largely independent of the microscopic details of individual interactions [1, 2]. When a sufficiently large number of agents interact according to local rules, collective behaviour emerges that cannot be inferred from any single agent in isolation. This phenomenon, termed *emergence*, finds its sharpest formal expression in the *phase transition*—a macroscopic reorganisation of a system driven by infinitesimal changes in a global control parameter. Such transitions are not confined to physical matter. Sharp collective reorganisations have been identified in models of opinion formation [3], residential sorting [4], technology adoption [5], and financial market dynamics [6], suggesting that the mathematical machinery of statistical mechanics—energy landscapes, order parameters, partition functions, and susceptibility divergences—constitutes a productive language for socioeconomic modelling [7, 8, 9].

Several structural features of socioeconomic systems align naturally with this formalism. Agents possess heterogeneous utilities that govern their disposition toward available states; transition rates between states follow Boltzmann-type expressions when idiosyncratic noise is present; and macroscopic observables—market shares, opinion distributions, wealth concentrations—play the role of thermodynamic order parameters whose dependence on global control parameters can be summarised in phase diagrams [3, 8]. The present work extends this programme to the domain of digital marketplaces, where two natural global controls—decision noise and information accessibility—are shown to organise an array of distinct market structures into a coherent phase diagram.

1.2 Discrete Choice, Boltzmann Weights, and Collective Market States

The connection between consumer choice theory and statistical mechanics is made precise through the discrete-choice framework introduced by McFadden [10]. When idiosyncratic preference shocks are drawn independently from a Gumbel distribution, the probability that agent i selects option j from a consideration set $\mathcal{C}_i$ takes the softmax (multinomial logit) form

$$\pi_{ij} \;=\; \frac{\exp\bigl(U_{ij}/T\bigr)}{\sum\limits_{k\in\mathcal{C}_i}\exp\bigl(U_{ik}/T\bigr)}, \tag{1}$$

where U_{ij} is the systematic utility of vendor j for customer i and

$T > 0$ is a scale parameter governing idiosyncratic noise. The structural equivalence of equation (1) with a Boltzmann distribution at temperature T is well established [11, 12]. At low T, agents act nearly deterministically, selecting the highest-utility option; at high T, choices approach the uniform distribution over $\mathcal{C}_i$. This single parameter therefore provides a theoretically grounded, continuously variable control for the degree of rationality—equivalently, the degree of thermal disorder—present in the market.

Social interactions introduce a second structural layer. When the utility of an option depends on how many other agents have already selected it, a self-referential fixed-point condition arises. Each vendor j is assigned a binary type $s_j \in \{-1, +1\}$, where $s_j = -1$ denotes a local vendor and $s_j = +1$ denotes a digital vendor. The market state is summarised by the signed order parameter

$$m \;=\; \sum_{j=1}^{M} s_j \, p_j, \tag{2}$$

where p_j is the equilibrium market share of vendor j and M is the total number of vendors. Adding a social coupling term $J s_j m$ to the utility function yields a mean-field structure formally isomorphic to the Curie–Weiss infinite-range Ising model [13, 14]. The parameter $J \geq 0$ tunes the strength of this coupling: at small J the market resides in a disordered, mixed phase; above a critical value $J^*(T)$ the symmetry is broken and the market tips toward dominance by one vendor type [15]. Hysteresis, bistability, and discontinuous transitions all arise naturally from this structure [16], providing a precise mathematical language for the tipping-point phenomena discussed in the platform-economics literature [18, 19]. Informational cascades driven by social influence [17] emerge as a special case of this framework when J is large relative to T.

A third control, largely absent from prior analytic treatments but central to the present model, governs the information available to each agent. In practice, no consumer evaluates the full vendor population; each agent's consideration set $\mathcal{C}_i$ is a strict subset of the market. This is parameterised by the visibility bandwidth $\gamma \in [0, 1]$, a continuous scalar that scales the per-round probability with which each customer discovers nearby local vendors: at $\gamma = 0$ no local vendor is ever discovered, while at $\gamma = 1$ the full distance-weighted discovery kernel is active. The pair (T, γ) thus provides two orthogonal global controls—noise and information—whose joint variation sweeps out a rich phase diagram independently of the social coupling J.

1.3 Agent-Based and Mean-Field Approaches to Market Dynamics

Two methodological traditions serve the programme outlined above. Agent-based models (ABMs) simulate explicit micro-level decisions over discrete time, allowing emergent macro-structure to be read directly from simulation trajectories [20, 21]. ABMs naturally accommodate heterogeneous agents, finite-population fluctuations, path dependence, and dynamic memory effects such as habit reinforcement; their principal limitation is computational, as systematic parameter sweeps across a high-dimensional control space become costly at large population sizes.

Mean-field approximations address this limitation by replacing the full heterogeneous interaction landscape with a self-consistency equation for the expected order parameter. The fixed-point problem

$$m \;=\; \mathcal{F}(m;\, T, \gamma, J) \tag{3}$$

is solved iteratively and, under mild regularity conditions ($T > 0$, bounded utilities, appropriate damping), converges to a unique or finitely many equilibria m^* [15, 16]. Equilibrium vendor shares $p_j^* = p_j(m^*)$ follow directly. The mean-field approach enables rapid, smooth exploration of the (T, γ) plane and yields clean phase boundaries locatable through susceptibility peaks

$$\chi_T \;=\; \frac{\partial L^*}{\partial T}, \qquad \chi_\gamma \;=\; \frac{\partial L^*}{\partial \gamma}, \tag{4}$$

where $L^* = \sum_{j:\, s_j = -1} p_j^*$ is the equilibrium local market share. Maxima in equations (4) signal phase boundaries in precisely the same sense that magnetic susceptibility diverges at the Curie temperature [1]. The two approaches are complementary rather than competing. A formal equivalence, arising from the law of large numbers, is established in Section 2: in the limit $N \to \infty$ and in the absence of long-lived memory transients, the time-averaged ABM trajectory converges almost surely to the mean-field fixed point. Deviations from this limit quantify finite-size effects, hysteresis, and slow mixing—phenomena that the ABM alone can resolve. The dual-engine design adopted here exploits both capabilities: the mean-field engine is used for rapid phase-diagram construction, while the ABM is used for selective validation and the study of transient and memory-driven dynamics.

1.4 Scope, Contributions, and Organisation

A unified model of customer–vendor choice on a bipartite network of N customers and M vendors is developed and analysed. Vendors are partitioned into two types—*local* (geographically proximate, distance-dependent delivery latency) and *digital* (platform-mediated, parametric latency)—and the market state is characterised by the signed or-

der parameter m of equation (2). Three qualitatively distinct market phases are identified: *Local Dominance* ($m \ll 0$), *Digital Dominance* ($m \gg 0$), and *Mixed Equilibrium* ($m \approx 0$). Phase boundaries are mapped in the (T, γ) plane for a range of social coupling strengths J.

Three specific contributions are made. First, a calibrated logistic visibility kernel is introduced that maps the continuous parameter γ to an interpretable per-round probability of discovering near versus far local vendors, anchored at the fifth and ninety-fifth percentiles of the empirical distance distribution; this grounding allows γ to be varied continuously without loss of physical interpretability. Second, two-dimensional phase diagrams are computed via the mean-field engine and validated against ABM realisations, revealing that the transition between Digital Dominance and Mixed Equilibrium is continuous (second-order-like) in T but can become discontinuous (first-order-like) at large J, with a corresponding hysteresis loop recoverable from the ABM. Third, the two engines are validated against each other across the full (T, γ) plane, revealing that they differ not only quantitatively but in which control parameter is dynamically active near the phase boundary: γ acts as a genuine dynamical control in the ABM through its accumulating-knowledge mechanism, whereas the mean-field static-mask approximation largely suppresses χ_γ, and a fine-grid susceptibility analysis identifies localised information-threshold hotspots absent from the mean-field solution.

The remainder of the paper is organised as follows. Section 2 presents the full model specification: the baseline utility function, the logistic visibility kernel, the social coupling term, the mean-field fixed-point engine, the ABM, and the formal law-of-large-numbers equivalence between them. Section 3 reports phase diagrams and susceptibility analyses across the (T, γ, J) parameter space. Section 4 discusses the results in relation to prior work on market tipping points and platform competition. Section 5 addresses limitations and directions for future work, and Section 6 concludes.

2. Methods

2.1 Population generation

All agents live on a bounded 2D geography, taken to be the unit square $[0, 1] \times [0, 1]$. Customers have explicit coordinates, while only *local* vendors have physical coordinates. Digital vendors are treated as spatially delocalized and are instead parameterized by a delivery/latency variable.

2.1.1 Customer population

Each customer $i \in \{1, \ldots, N\}$ is assigned a location $\mathbf{x}_i = (x_i, y_i)$ and four heterogeneous preference parameters: price sensitivity α_i, delay sensitivity u_i, reinforcement strength δ_i, and a social-noise amplitude s_i.

Customer locations are sampled from a clipped Gaussian centered at the market center:

$$x_i \sim \mathrm{clip}\big(\mathcal{N}(0.5,\, 0.2^2),\, 0,\, 1\big), \qquad y_i \sim \mathrm{clip}\big(\mathcal{N}(0.5,\, 0.2^2),\, 0,\, 1\big). \tag{5}$$

To encode a time–money tradeoff, (α_i, u_i) are sampled jointly from a bivariate normal with *negative* correlation:

$$\begin{pmatrix}\alpha_i \\ u_i\end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix}\mu_\alpha \\ \mu_u\end{pmatrix}, \begin{pmatrix}\sigma_\alpha^2 & \rho_{\alpha u}\sigma_\alpha\sigma_u \\ \rho_{\alpha u}\sigma_\alpha\sigma_u & \sigma_u^2\end{pmatrix}\right), \qquad (\alpha_i, u_i) \leftarrow \mathrm{clip}\big((\alpha_i, u_i),\, 0.05,\, 3.0\big), \tag{6}$$

with default parameters $\mu_\alpha = \mu_u = 1.0$, $\sigma_\alpha = \sigma_u = 0.30$, and $\rho_{\alpha u} = -0.5$. Reinforcement δ_i and social-noise amplitude s_i are sampled independently and clipped to ensure physical ranges:

$$\delta_i \sim \mathrm{clip}\big(\mathcal{N}(\mu_\delta, \sigma_\delta^2),\, 0.05,\, 3.0\big), \qquad s_i \sim \mathrm{clip}\big(\mathcal{N}(\mu_s, \sigma_s^2),\, 0,\, 2.0\big), \tag{7}$$

with defaults $\mu_\delta = 1.0, \sigma_\delta = 0.30$ and $\mu_s = 0.30, \sigma_s = 0.10$.

2.1.2 Vendor population

We generate M vendors, partitioned into M_{L} local vendors and $M_{\mathrm{D}} = M - M_{\mathrm{L}}$ digital vendors, where $M_{\mathrm{L}} = \mathrm{round}(f_{\mathrm{L}} M)$ and $f_{\mathrm{L}} \in [0, 1]$ is the local fraction. Local vendors are assigned spatial coordinates $\mathbf{y}_j = (x_j, y_j)$ uniformly on the unit square:

$$(x_j, y_j) \sim \mathrm{Uniform}\big([0,1] \times [0,1]\big) \quad \text{for local vendors.} \tag{8}$$

Digital vendors have no physical coordinates in the data representation and are instead parameterized by a delivery-delay variable τ_j:

$$\tau_j \sim \max\big(0.1,\, \mathcal{N}\big(\rho\, d_{\mathrm{ref}},\, \sigma^2\big)\big) \quad \text{for digital vendors,} \tag{9}$$

where $d_{\mathrm{ref}} = 0.521$ is a fixed reference scale used in the implementation, ρ sets the mean digital delay, and σ sets the delay heterogeneity.

Each vendor also receives basic economic attributes. Operational cost c_j and margin m_j are sampled from clipped Gaussians, and the posted price is $p_j = c_j + m_j$ (with an optional final clip):

$$\begin{aligned} &c_j \sim \mathrm{clip}\big(\mathcal{N}(\mu_c, \sigma_c^2),\, c_{\min}, c_{\max}\big), \qquad m_j \sim \mathrm{clip}\big(\mathcal{N}(\mu_m, \sigma_m^2),\, m_{\min}, m_{\max}\big), \\ &p_j \leftarrow \mathrm{clip}(c_j + m_j,\, p_{\min}, p_{\max}). \end{aligned} \tag{10}$$

Finally, the rating is generated as a noisy (typically decreasing) function of price:

$$r_j = \mathrm{clip}\Big(r_0 + \beta_p\, p_j + q_j + \epsilon_j,\ 1,\ 5\Big), \tag{11}$$

where $\epsilon_j \sim \mathcal{N}(0, \sigma_r^2)$ and q_j is an optional independent "quality" term (disabled by default).

2.1.3 Population reuse across parameter sweeps

To ensure that phase sweeps vary only the *control parameters* (e.g., T, γ) and not the underlying market realization, the simulation driver caches each population scenario keyed by the generation parameters $(N, M, f_{\mathrm{L}}, \rho, \sigma)$ and reuses it across repeated runs and across the sweep grid.

2.2 Population diagnostics

Figures 1–5 summarize the realized population for the primary parameter setting ($N = 200$, $M = 40$, $\lambda = 0.5$). Figure 1 shows the spatial arrangement of customers and local vendors in the unit square; digital vendors are delocalized and are therefore not assigned positions. Figures 2 and 3 display the marginal and joint distributions of the four customer heterogeneity parameters, confirming the imposed negative correlation $\rho_{\alpha u} = -0.5$ between price sensitivity and delay sensitivity. Figures 4 and 5 show the vendor attribute distributions and the price–rating relationship, respectively. The negative slope in Figure 5 reflects the parameterisation in equation (11): higher-priced vendors receive lower average ratings under the default price-only rating model.

2.3 Utility function and nondimensionalisation

The systematic utility of vendor j for customer i at time t is a linear combination of economic, geographic, social, and memory terms:

$$U_{ij}^{(t)} \;=\; -\alpha_i\,\tilde{p}_j + \tilde{r}_j - u_i\,\tilde{d}_{ij} + \delta_i\,\log\!\big(1 + A_{ij}^{(t)}\big) + \phi\,\log\!\big(1 + n_j^{(t-1)}\big) + J\,s_j\,m, \tag{12}$$

where tildes denote population z-scores: for any raw feature x, $\tilde{x} = (x - \bar{x})/\sigma_x$, with mean $\bar{x}$ and standard deviation σ_x estimated jointly over all $N \times M$ customer–vendor pairs in the realized population. This nondimensionalisation ensures that the coefficients α_i, u_i, and 1 (on the rating term) are comparable in magnitude and that the temperature T acts as a single unified noise scale.

The effective delay $\tilde{d}_{ij}$ is computed differently for the two vendor types. For local vendors, the raw delay is the Euclidean travel distance $d_{ij} = \|\mathbf{x}_i - \mathbf{y}_j\|_2$ rescaled to the same units as τ_j via a population-level factor $\kappa = \bar{\tau}_{\mathrm{digital}}/\bar{d}_{\mathrm{local}}$, giving $d_{ij}^{\mathrm{eff}} = \kappa d_{ij}$. For digital vendors, the raw delay is τ_j. Both effective delays are then z-scored into $\tilde{d}_{ij}$.

The remaining terms in equation (12) are as follows. The affinity $A_{ij}^{(t)}$ is the cumulative count of times customer i has chosen vendor j up to round t; the logarithm with coefficient δ_i implements diminishing-returns habit reinforcement. The bandwagon term $\phi \log(1 + n_j^{(t-1)})$

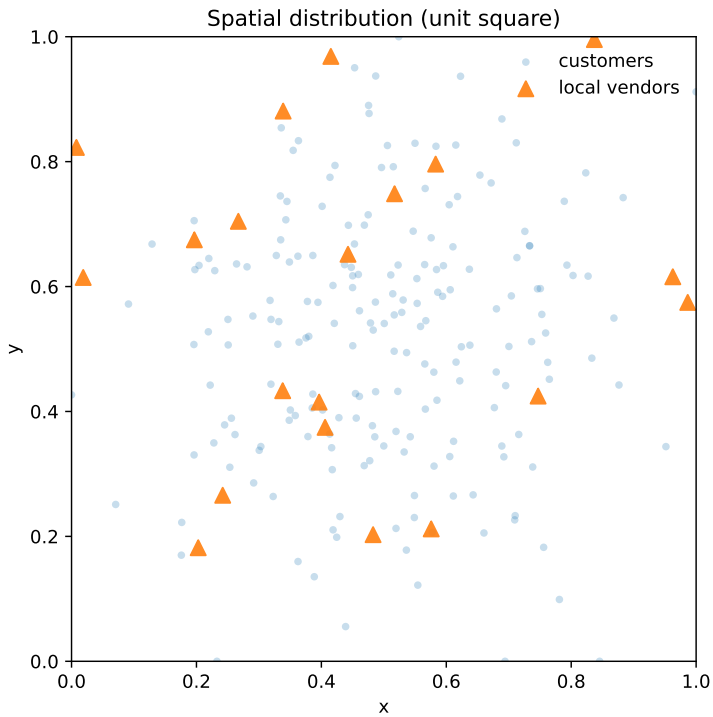


Figure 1. Spatial distribution of customers (points) and local vendors (triangles) in the unit-square geography. Digital vendors are delocalized and therefore not shown as spatial points.

adds a network-effect utility proportional to the log-count $n_j^{(t-1)}$ of customers who chose vendor j in the previous round, capturing popularity-begets-popularity dynamics at the aggregate level; ϕ and J are mechanistically distinct parameters—ϕ creates a round-level popularity effect weighted by recent counts, while J creates a mean-field interaction with the global magnetisation m. The social coupling term Js_jm is the mean-field interaction introduced in equation (2); $s_j \in \{-1, +1\}$ carries the sign of the vendor's type and m is the current order parameter. In the ABM, m is updated each round from the realized choices; in the mean-field engine, it is treated as a self-consistent parameter to be solved for (see Section 2.6).

2.4 Logistic visibility kernel

No customer evaluates all M vendors simultaneously. Each round, customer i's consideration set $\mathcal{C}_i^{(t)}$ is built by the following two rules. Digital vendors are always included: $\mathcal{C}_i^{(t)}$ contains every digital vendor

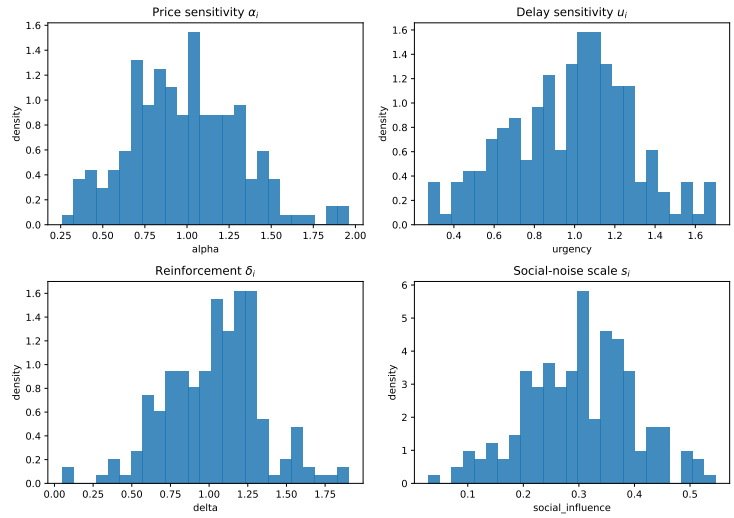


Figure 2. Customer heterogeneity distributions for price sensitivity α_i, delay sensitivity u_i, reinforcement δ_i, and social-noise scale s_i.

in every round, regardless of distance. Local vendors are discovered sequentially: at each round t, the probability that customer i first discovers local vendor j (given that j has not yet been discovered) is

$$P_{\text{disc}}(d_{ij};\,\gamma) \;=\; \gamma \cdot \sigma\Big(\beta\,(d_c - d_{ij})\Big), \qquad \sigma(z) = \frac{1}{1+e^{-z}}, \tag{13}$$

where β and d_c are population-calibrated parameters determined by the two anchor conditions $P_{\text{disc}}(d_5;\,\gamma=1) = 0.95$ and $P_{\text{disc}}(d_{95};\,\gamma = 1) = 0.05$, with d_5 and d_{95} denoting the fifth and ninety-fifth empirical percentiles of the full distance matrix $\{d_{ij}\}$. Once vendor j enters $\mathcal{C}_i^{(t)}$, it remains there for all subsequent rounds (accumulating-knowledge semantics). The scalar $\gamma \in [0,1]$ scales the overall discovery rate: at $\gamma = 0$ no local vendor is ever discovered, while at $\gamma = 1$ the full logistic kernel is active. This construction gives γ a concrete physical interpretation as an information-access bandwidth that can be varied continuously without disturbing the shape or calibration of the underlying distance–probability curve.

2.5 Agent-based model

The ABM advances the market through $T_{\max}$ discrete rounds. At the start of each run, affinities $A_{ij}^{(0)} = 0$ and the knowledge set $\mathcal{K}_i^{(0)} = \emptyset$ for all (i,j). Each round t proceeds in five steps.

Step 1: Visibility update. The accumulated-knowledge mask $\mathcal{K}_i^{(t)}$ is updated by drawing Bernoulli trials with probabilities given by equa-

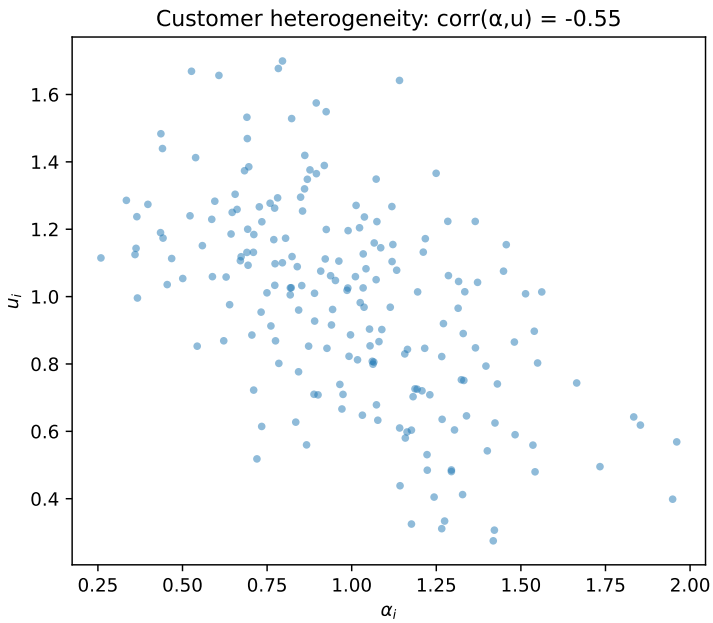


Figure 3. Joint distribution of (α_i, u_i) showing the imposed time–money trade-off as a negative correlation.

tion (13) for all undiscovered local vendors, and then taking the union with the previous mask: $\mathcal{K}_i^{(t)} = \mathcal{K}_i^{(t-1)} \cup \mathcal{D}_i^{(t)}$, where $\mathcal{D}_i^{(t)}$ denotes the set of newly discovered vendors. The consideration set is $\mathcal{C}_i^{(t)} = \mathcal{K}_i^{(t)}$.

Step 2: Utility computation. The utility matrix $U_{ij}^{(t)}$ is computed from equation (12) for all (i, j) pairs, using the current affinities $A_{ij}^{(t)}$, the previous-round counts $n_j^{(t-1)}$, and the current order parameter $m^{(t)} = \sum_j s_j \hat{p}_j^{(t-1)}$ computed from the empirical shares of the previous round.

Step 3: Masked softmax. For each customer i, a probability vector over vendors is computed by applying the softmax of equation (1) restricted to $\mathcal{C}_i^{(t)}$, with utilities divided by temperature T. Vendors outside the consideration set receive zero probability.

Step 4: Choice draw. Each customer independently draws one vendor from their probability vector via a vectorised categorical sample.

Step 5: Affinity update. The affinity of each chosen pair is incremented: $A_{ij}^{(t+1)} = A_{ij}^{(t)} + \mathbf{1}[\text{customer } i \text{ chose vendor } j]$. The round-$t$ vendor counts $n_j^{(t)}$ are stored for use in the next round's bandwagon term.

After $T_{\max} = 40$ rounds, the steady-state order parameters L^*,

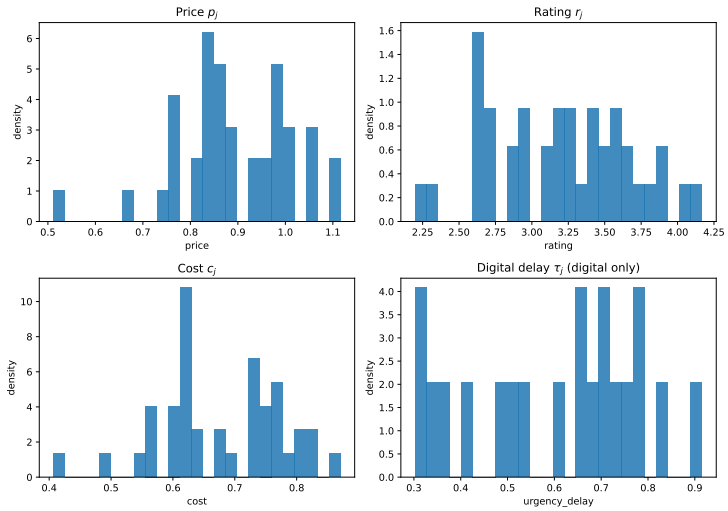


Figure 4. Vendor attribute distributions (price p_j, rating r_j, cost c_j, and digital delay τ_j for digital vendors only).

HHI*, and H^* are estimated as averages over the final five rounds.

2.6 Mean-field fixed-point engine

The mean-field engine replaces the stochastic choice trajectories of the ABM with a deterministic self-consistency equation for the order parameter m^*. The computation proceeds in three stages.

Stage 1: Base-utility precomputation. The population-level z-score statistics are estimated from the realized customer and vendor data. For each pair (i, j), the baseline utility $U_{ij}^{(0)} = -\alpha_i \tilde{p}_j + \tilde{r}_j - u_i \tilde{d}_{ij}$ is computed, omitting the affinity, bandwagon, and social-coupling terms. These terms are either absent (affinities initialise to zero and are not iterated in the mean-field engine) or folded into the self-consistency equation.

Stage 2: Visibility freeze. The accumulating-knowledge process of Section 2.4 is simulated for $T_{\max}$ rounds with a fixed random seed to produce a frozen knowledge mask $\overline{\mathcal{K}}_i$ for each customer. This approximates the steady-state consideration set that would be reached in the ABM without requiring the full round-by-round simulation.

Stage 3: Fixed-point iteration. Starting from an initial estimate $m^{(0)}$, the iteration

$$m^{(k+1)} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \in \overline{\mathcal{C}}_i} s_j \frac{\exp\big[(U_{ij}^{(0)} + J s_j m^{(k)})/T\big]}{\sum_{l \in \overline{\mathcal{C}}_i} \exp\big[(U_{il}^{(0)} + J s_l m^{(k)})/T\big]} \tag{14}$$

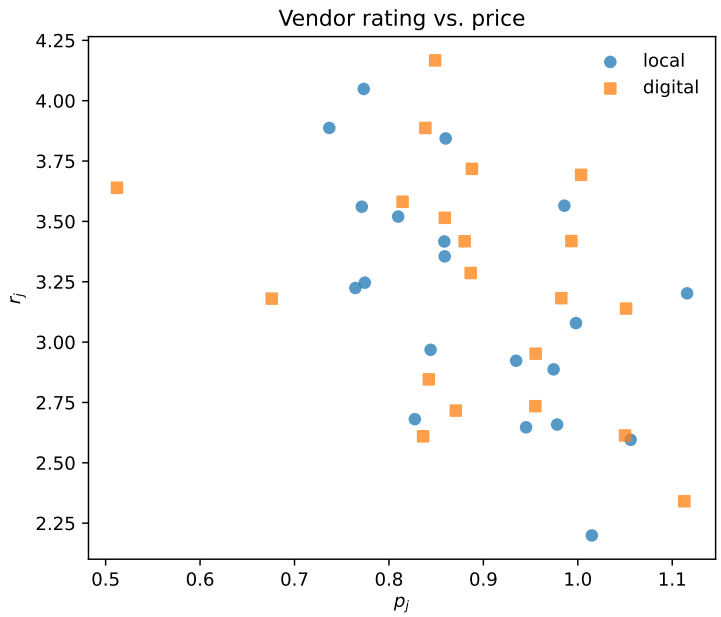


Figure 5. Vendor rating vs. price, separated by vendor type.

is applied with damping until $|m^{(k+1)} - m^{(k)}| < 10^{-6}$ or a maximum iteration count is reached. Convergence is declared at the resulting m^*, and equilibrium vendor shares $p_j^* = N^{-1} \sum_i \pi_{ij}(m^*)$ are read off directly.

2.7 Formal equivalence in the large-population limit

The ABM and mean-field engine are connected by a law-of-large-numbers argument. At any round t, the empirical market share of vendor j is $\hat{p}_j^{(t)} = N^{-1} \sum_i \mathbf{1}[\text{customer } i \text{ chose } j]$. Because choices are conditionally independent given the utility matrix and temperature, the weak law of large numbers gives $\hat{p}_j^{(t)} \xrightarrow{p} \mathbb{E}[\pi_{ij}^{(t)}]$ as $N \to \infty$, where the expectation is over the heterogeneous customer population. In the stationary regime, where affinities and bandwagon counts have equilibrated, this expected share satisfies exactly the fixed-point equation (14) with $m^{(k)} = m^{(k+1)} = m^*$. Thus the mean-field fixed point is the almost-sure limit of the ABM's steady-state empirical distribution as $N \to \infty$. At the finite population size $N = 200$ used here, deviations between the two engines scale as $\mathcal{O}(N^{-1/2})$ in the bulk of each phase; near phase boundaries, where susceptibility is large, fluctuations are amplified and the ABM may exhibit bistability absent from the mean-field solution (see Section 3.8).

3. Results

We report results from a Cartesian temperature–visibility sweep at fixed market structure and social coupling, comparing the stochastic agent-based model (ABM) against the deterministic mean-field fixed-point engine across the full (T, γ) control plane. Sections 3.2–3.3 establish the steady-state methodology and the primary phase diagram. Section 3.4 refines the phase boundary at higher grid resolution. Section 3.5 characterises the temperature axis in detail via a targeted 1D slice at fixed γ. Sections 3.6–3.7 characterise the secondary order parameters and locate phase boundaries quantitatively. Sections 3.8–3.9 validate the dual-engine design and examine the role of social coupling. A summary of all three phases is given in Section 3.10.

3.1 Simulation Design and Parameter Space

All results are obtained from a Cartesian sweep over decision temperature $T \in [0.1, 2.0]$ and information bandwidth $\gamma \in [0.0, 1.0]$, each sampled at 21 uniformly spaced values, yielding a 441-point grid. At every grid point both the mean-field fixed-point engine and the ABM are executed and stored separately. Social coupling is fixed at $\phi = J = 0.2$ throughout the primary sweep (where ϕ is the bandwagon coupling and J is the Ising-type mean-field coupling, set equal and varied jointly here); sensitivity to J is examined in Section 3.9.

The market population consists of $N = 200$ customers and $M = 40$ vendors, with local fraction $\lambda = 0.5$, giving 20 local and 20 digital vendors. Vendor economics are generated with cost parameters $\mu_c = 0.70$, $\sigma_c = 0.10$, and margin parameters $\mu_m = 0.20$, $\sigma_m = 0.05$. Digital delivery delays are drawn as $\tau_j \sim \max(0.1, \mathcal{N}(\rho\bar{d}, \sigma_\tau))$ with $\rho = 1.2$, $\sigma_\tau = 0.20$, and $\bar{d} \approx 0.521$, the expected Euclidean distance between two uniformly distributed points in the unit square. This calibration places the mean digital latency at $\rho = 1.2$ times the mean local travel distance, giving the two vendor types comparable but non-identical competitive positions at full visibility $\gamma = 1$. Each parameter combination is replicated $R = 10$ times with independent random seeds, and all reported order parameters are averaged across replicates. A rolling late-window average over the final five simulation rounds serves as the steady-state estimator for all ABM metrics, a choice justified by the equilibration diagnostics presented below.

3.2 Equilibration and Convergence

Figure 6 presents a representative time series of the three primary order parameters across 40 simulation rounds. Local market share $L^{(t)}$, HHI, and Shannon entropy $H^{(t)}$ each exhibit rapid transients in the first 10–15 rounds, driven by the initial accumulation of local-vendor

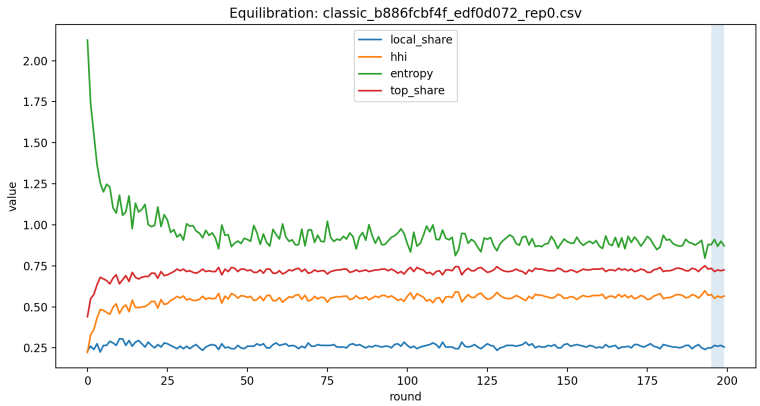


Figure 6. Equilibration diagnostics for a representative ABM run ($T = 0.5$, $\gamma = 0.5$, $\phi = 0.2$, $N = 200$, $M = 40$). Local market share $L^{(t)}$, Herfindahl–Hirschman Index $\mathrm{HHI}^{(t)}$, and Shannon entropy $H^{(t)}$ are plotted against simulation round. All three metrics stabilise within approximately 20 rounds, well before the late-window averaging interval (rounds 36–40), indicated by the grey shading.

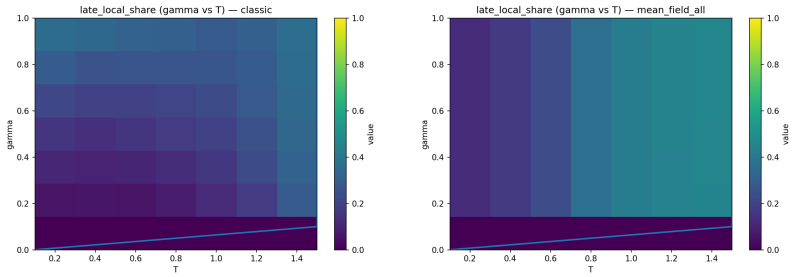
knowledge through the calibrated logistic visibility kernel, before settling to stable plateaux. This confirms that the late-window averaging interval (rounds 36–40) provides a reliable steady-state proxy across all market configurations tested. For the mean-field engine, convergence is assessed via the fixed-point residual $|m^{(k+1)} - m^{(k)}|$, which falls below 10^{-6} within 50 iterations for every (T, γ) point in the sweep.

3.3 Phase Diagram in the (T, γ) Plane

The central result is the phase diagram in Figure 7, mapping steady-state local market share L^* across the full (T, γ) plane. Three qualitatively distinct regimes are identified in terms of the magnetization order parameter m^* of equation (2).

Digital Dominance ($m^* > 0$, $L^* < 0.5$) occupies the lower-left region, corresponding to low decision noise and poor information access. At small γ, the calibrated logistic kernel strongly suppresses local-vendor discovery: the per-round probability $P_{\mathrm{disc}} = \gamma \cdot \sigma(\beta(d_c - d_{ij}))$ is small because γ is small, so digital vendors—always visible by construction—command nearly all market share. Low temperature then concentrates choices deterministically on the visible options, producing an almost entirely digital equilibrium.

Local Dominance ($m^* < 0$, $L^* > 0.5$) occupies the upper-right region, where γ is large enough for customers to accumulate knowledge of nearby local vendors across the simulation horizon. Once a local ven-



(a) Classic (ABM) simulation. (b) Mean-field fixed point.

Figure 7. Steady-state local market share L^* over the (T, γ) control plane. Both engines reveal a low-γ regime of Digital Dominance and a high-γ regime of Local Dominance, separated by a crossover whose width differs markedly between the two approaches: the mean-field boundary is sharper and more vertical, while the ABM crossover is broadened by stochastic and memory-driven effects. The dashed contour marks $L^* = 0.5$.

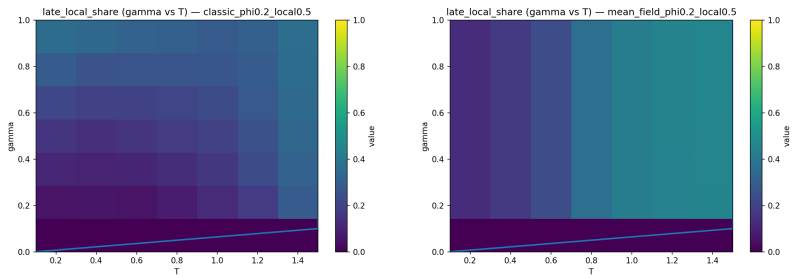
dor enters a customer's consideration set, its competitive advantage—arising from the short Euclidean distance d_{ij} and hence small urgency-delay penalty $\upsilon_i d_{ij}$—appears as a utility premium over digital vendors, whose mean delay $\rho\bar{d} \approx 0.625$ exceeds the delay of any local vendor within roughly the median customer–vendor distance. The social coupling term $Js_j m^*$ amplifies this advantage via positive feedback: growing local share makes m^* more negative, which further raises local utility through equation (2).

Mixed Equilibrium ($m^* \approx 0$) is found at high T across all γ. When decision noise dominates, the softmax probabilities of equation (1) flatten toward the uniform distribution, erasing utility gradients between vendor types. Market share then distributes approximately in proportion to the size of each type's consideration set; because $\lambda = 0.5$, the result is $L^* \approx 0.5$ with near-maximal Shannon entropy.

3.4 Fine-Grid Boundary Survey: Zoomed (T, γ) Sweep

To resolve the phase boundary at higher resolution, we performed a second sweep confined to the boundary region identified in Figure 7, using a finer 9×9 grid: $T \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50, 0.60, 0.70\}$ and $\gamma \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50\}$, with 200 rounds per run, $R = 10$ replicates, and all other parameters held at their primary-sweep values ($\phi = J = 0.2$, $\lambda = 0.5$, $N = 200$, $M = 40$).

Local adoption surface. In the zoomed ABM map (Figure 8, left), local market share L^* rises smoothly with both T and γ across the full grid, with no sharp jump visible at this resolution. This is consistent with a broadened crossover rather than a first-order transition



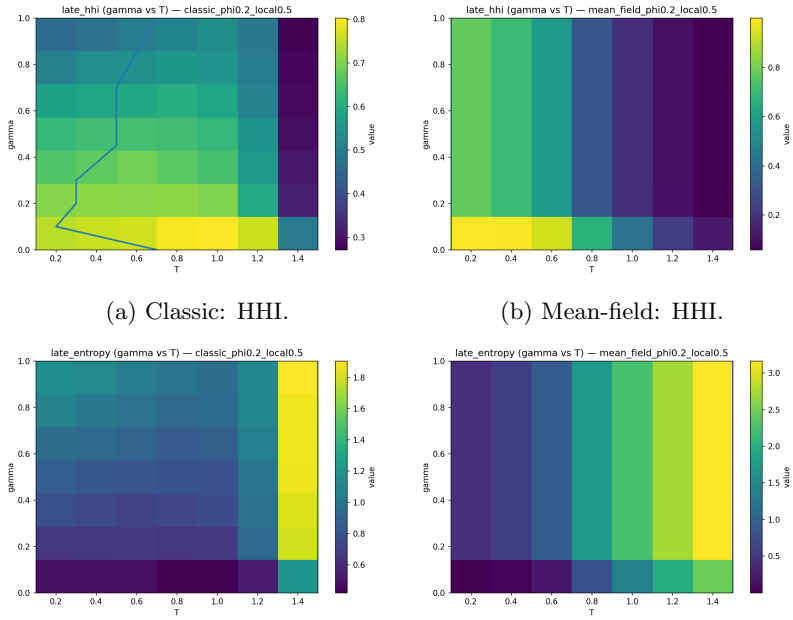
(a) Classic (ABM). (b) Mean-field fixed point.

Figure 8. Steady-state local market share L^* in the zoomed (T, γ) plane ($T \in [0.10, 0.70]$, $\gamma \in [0.10, 0.50]$, 9×9 grid, 200 rounds, $R = 10$ replicates). Both axes are sampled at finer resolution than the coarse sweep of Figure 7, resolving the gradient near the boundary between Digital Dominance and Local Dominance. In the ABM (left), local share increases smoothly with both T and γ. In the mean-field engine (right), the response is dominated by T and the γ axis is substantially compressed.

in the ABM at $J = 0.2$: the stochastic, reinforcement-driven dynamics produce a gradual slope rather than a discontinuous boundary. In the zoomed mean-field map (Figure 8, right), the response is predominantly organised along the T axis; the γ dependence is compressed relative to the ABM. This divergence reflects the fundamental modelling difference discussed in Section 3.8: the static-mask mean-field approximation fixes each customer's consideration set and therefore loses the round-by-round discovery dynamics that make γ an active control parameter in the ABM.

Concentration and diversity. The zoomed HHI and entropy maps (Figure 9) confirm that the ABM responds to γ as well as T throughout this region. In the ABM, HHI declines and entropy rises with increasing γ at every temperature tested, demonstrating that improved information access consistently diversifies the market even when the transition in local share is gradual. In the mean-field maps the same γ dependence is substantially damped, and the dominant gradient runs along T alone; this is the analogue of the "Boltzmann thermalisation" signal already observed in the coarse sweep.

Sensitivity maps and the information threshold. The zoomed susceptibility maps (Figure 10) reveal structure that is obscured in the coarse 7-point grid. In the ABM, $\chi_\gamma = \partial L^*/\partial\gamma$ shows localised hotspots rather than a uniform ridge, indicating that the marginal effect of information access is strongest in specific cells rather than globally. These pockets are consistent with a threshold phenomenon: once enough local vendors enter customer consideration sets, additional informa-



(a) Classic: HHI.

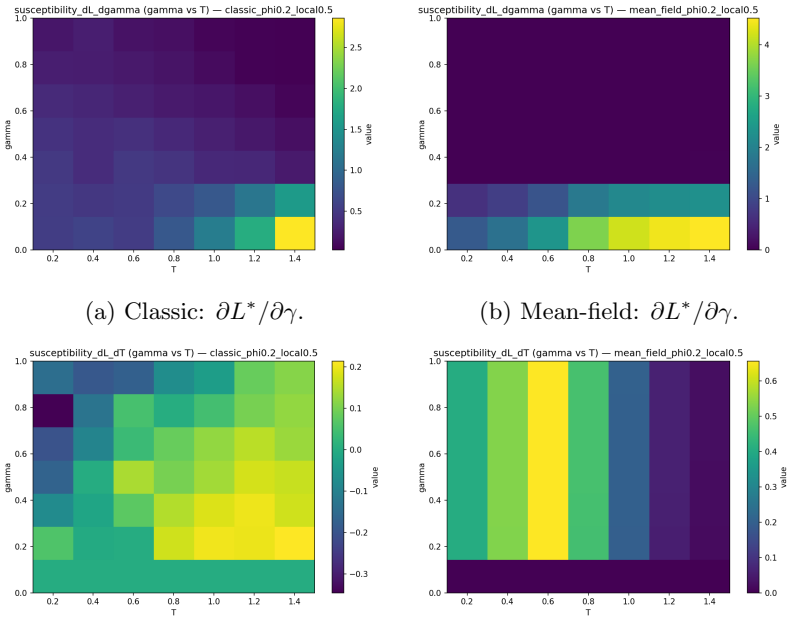
(b) Mean-field: HHI.

(c) Classic: Shannon entropy.

(d) Mean-field: Shannon entropy.

Figure 9. Concentration (HHI, top row) and diversity (Shannon entropy, bottom row) over the zoomed (T, γ) plane, for the ABM (left column) and mean-field engine (right column). In the ABM, both metrics respond to γ as well as T: increasing information access reduces HHI and raises entropy, confirming that the visibility mechanism operates as an active dynamical control parameter in this region. In mean-field, the γ axis is substantially damped, and the dominant gradient runs along T alone.

tion translates into measurable adoption, while outside that range the marginal effect is smaller. The ABM χ_T map is patchier, reflecting that finite-sample noise in the 5-round late-window estimator amplifies under differentiation and that temperature effects are more gradual than visibility effects in this parameter region. In the mean-field engine, χ_T displays a coherent broad ridge at intermediate T, while χ_γ is near-zero over most of the grid. The suppression of χ_γ in mean-field is particularly diagnostic: it confirms that the static-mask approximation is not reproducing the ABM's sensitivity to information access, and that the two engines differ not only quantitatively but also in which control parameter is most active within this region.



(a) Classic: $\partial L^*/\partial\gamma$.

(b) Mean-field: $\partial L^*/\partial\gamma$.

(c) Classic: $\partial L^*/\partial T$.

(d) Mean-field: $\partial L^*/\partial T$.

Figure 10. Finite-difference susceptibility maps in the zoomed (T, γ) plane. Top row: $\chi_\gamma = \partial L^*/\partial\gamma$. Bottom row: $\chi_T = \partial L^*/\partial T$. In the ABM (left column), χ_γ shows localised hotspots consistent with a visibility threshold: marginal increases in information access produce the largest gains in local share at specific (T, γ) cells rather than uniformly across the grid. The ABM χ_T map is patchier, reflecting both the stochastic sampling noise in the late-window averages and the more gradual temperature effect in this parameter region. In mean-field (right column), χ_T exhibits a coherent broad ridge at intermediate T corresponding to the most responsive mixing band, while χ_γ is near-zero across the grid—confirming that the static-mask approximation suppresses the information-access axis relative to the full ABM dynamics.

3.5 Temperature Response at Fixed Information Access ($\gamma = 0.3$)

We first isolate the role of decision noise by sweeping temperature at fixed information access. In this preliminary run the sweep parameters were $T \in \{0.1, 0.5, 1.0\}$ and $\gamma = 0.3$, with $\phi = 0.2$ in the classic ABM and $J = 0.2$ in the mean-field model, market size $N = 200$ customers and $M = 40$ vendors, local fraction 0.5, and 40 rounds per simulation. Results are averaged over $R = 5$ independent replicates ($\rho = 1.2$, $\sigma = 0.2$). Summary metrics are computed as (i) `final_*` from the last round and (ii) `late_*` as an average over the last five rounds.

Local market share increases with temperature, but ABM and mean-field disagree on magnitude. At $\gamma = 0.3$, the classic ABM remains digital-dominant across the tested range: the late local share rises from 0.147 ± 0.011 at $T = 0.1$ to 0.248 ± 0.031 at $T = 1.0$. In contrast, the mean-field approximation predicts a much stronger shift toward local purchasing: 0.129 ± 0.000 at $T = 0.1$, 0.364 ± 0.000 at $T = 0.5$, and 0.441 ± 0.000 at $T = 1.0$. The mean-field model therefore moves the system toward a mixed (near-parity) regime by $T = 1.0$, while the ABM does not.

Market concentration decreases with temperature; mean-field strongly underestimates concentration at moderate and high T. In the ABM, concentration remains high even as T increases: late HHI decreases from 0.633 ± 0.045 ($T = 0.1$) to 0.485 ± 0.020 ($T = 1.0$), and the late top-vendor share stays large (0.794 ± 0.032 at $T = 0.1$; 0.674 ± 0.045 at $T = 1.0$). The mean-field model predicts substantially more dispersed outcomes at higher temperature: late HHI drops from 0.773 ± 0.000 ($T = 0.1$) to 0.104 ± 0.000 ($T = 1.0$), and late top share from 0.867 ± 0.000 to 0.241 ± 0.000. This is a qualitative ABM–mean-field divergence: the ABM sustains strong winner-take-most structure at moderate and high noise, while mean-field rapidly thermalises the market.

Entropy increases with temperature, with a large ABM–mean-field gap at $T \geq 0.5$. Consistent with the HHI and top-share trends, late entropy rises with T in both models but saturates much earlier in the ABM. In the ABM, late entropy increases from 0.510 ± 0.075 ($T = 0.1$) to 1.066 ± 0.193 ($T = 1.0$). In mean-field, late entropy is much larger at moderate and high T: 0.420 ± 0.000 ($T = 0.1$), 1.704 ± 0.000 ($T = 0.5$), and 2.733 ± 0.000 ($T = 1.0$). Interpreting entropy as the effective breadth of vendor choice, mean-field predicts far broader consideration and selection at $T \geq 0.5$ than is realised in the finite-agent ABM.

Temperature susceptibility highlights where local share is most sensitive to noise. We estimate the susceptibility $\chi_T = \partial L / \partial T$ using finite differences on the `late_local_share` curve (reported in Table 2). In the ABM, susceptibility is moderate and increases with T across this coarse grid: $\chi_T = 0.118 \pm 0.071$ at $T = 0.1$, 0.127 ± 0.025 at $T = 0.5$, and 0.132 ± 0.072 at $T = 1.0$. In mean-field, susceptibility is largest at low temperature and decreases with T: $\chi_T = 0.588 \pm 0.000$ at $T = 0.1$, 0.395 ± 0.000 at $T = 0.5$, and 0.154 ± 0.000 at $T = 1.0$. With only three temperature points these results support a smooth crossover rather than a sharply localised critical ridge at $\gamma = 0.3$; however, the

ABM–mean-field discrepancy in both the level of L and the magnitude and location of χ_T is robust within this sweep.

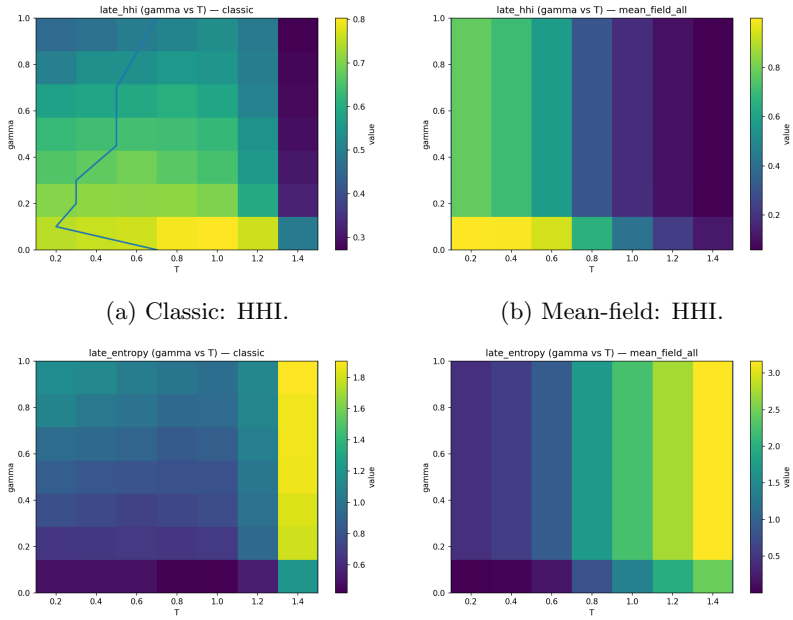
Equilibration and stability within the measured horizon. Mean-field fixed-point iteration converged in all runs with iteration counts decreasing with temperature (mean iterations: 30 at $T = 0.1$, 29 at $T = 0.5$, 22 at $T = 1.0$). In the ABM, the difference between `final` and `late` metrics is small at low and intermediate temperature (mean $|L_{\text{final}} - L_{\text{late}}| = 0.0005$ at $T = 0.1$, 0.0011 at $T = 0.5$) but larger at $T = 1.0$ (mean 0.0206), indicating stronger end-of-run fluctuations at higher noise within a 40-round horizon. Consequently, ABM uncertainty bands should be reported from replicate variability rather than from single-run final values.

Key takeaway. At fixed information access ($\gamma = 0.3$), increasing temperature systematically increases local share and reduces concentration. The mean-field approximation captures these monotonic trends but substantially overpredicts market dispersion and the shift toward local purchasing at moderate and high temperature relative to the classic ABM under the same parameterisation. Full numerical values are compiled in Tables 1–2.

3.6 Entropy and Market Concentration

Figure 11 shows the joint behaviour of Shannon entropy $H^* = -\sum_j p_j^* \log p_j^*$ and the Herfindahl–Hirschman Index $\text{HHI}^* = \sum_j (p_j^*)^2$ across the phase diagram. In the Mixed-Equilibrium regime, H^* approaches its theoretical maximum $\log M$ and HHI^* approaches its minimum $1/M$, consistent with near-uniform vendor shares. In the ordered phases, entropy falls sharply and HHI rises, indicating concentration among the dominant vendor type.

The steepest HHI gradients appear near the Digital-Dominance boundary at low γ. This concentration is structural rather than social-coupling-driven: when consideration sets are severely restricted, the few visible digital vendors compete within each customer's narrow window, and at low T a single vendor with marginally higher utility draws nearly all probability mass. The result is high HHI even before the social coupling term has time to act—an information-scarcity effect entirely distinct from the bandwagon-driven concentration seen at larger J (see Section 3.9). Temperature further modulates this structure: higher T increases diversity and reduces concentration throughout, but the ABM exhibits a richer, non-monotone response in the interior of the (T, γ) plane than the smoother mean-field prediction.



(a) Classic: HHI.

(b) Mean-field: HHI.

(c) Classic: Shannon entropy.

(d) Mean-field: Shannon entropy.

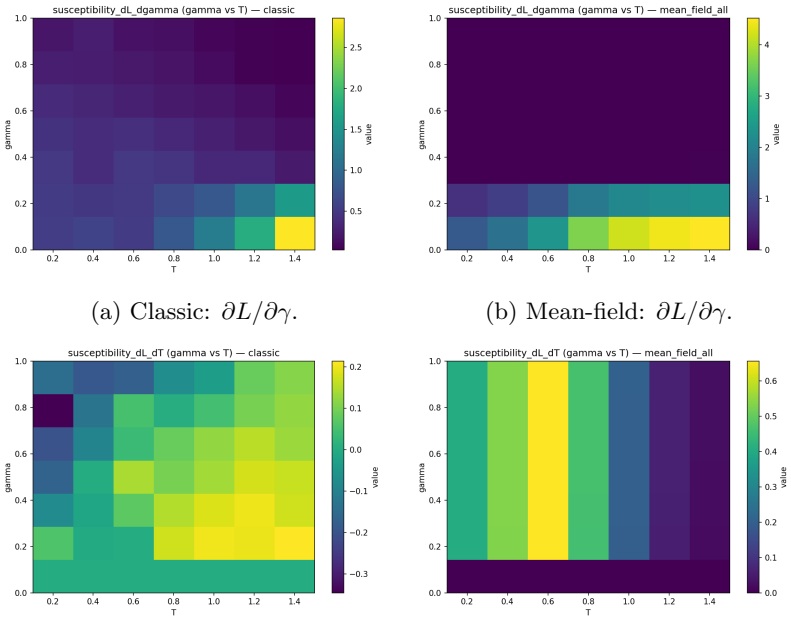
Figure 11. Herfindahl–Hirschman Index (top row) and Shannon entropy (bottom row) over the (T, γ) plane, for the ABM (left column) and mean-field engine (right column). HHI and entropy move in opposite directions across every phase boundary: the low-γ, low-T Digital Dominance regime is maximally concentrated and minimally diverse, while the high-T Mixed Equilibrium is minimally concentrated and maximally diverse. Mean-field reproduces the overall structure but compresses the transition and smooths away the non-monotone interior heterogeneity visible in the ABM maps.

3.7 Susceptibility Analysis and Phase-Boundary Location

Phase boundaries are located via the susceptibility functions $\chi_T = \partial L^*/\partial T$ and $\chi_\gamma = \partial L^*/\partial \gamma$ of equation (4), computed numerically from the 21×21 array of mean-field L^* values. The resulting surfaces are shown in Figure 12.

The χ_γ surface (top row) exhibits a pronounced ridge of large $|\chi_\gamma|$ at intermediate T values, locating the critical bandwidth $\gamma^*(T)$ above which the market tips from Digital to Local Dominance. At low T this ridge is narrow and intense, consistent with a sharp, potentially discontinuous transition reinforced by social coupling. At high T the ridge dissipates as both ordered phases are thermally suppressed.

The χ_T surface (bottom row) shows a ridge extending across all γ, tracking the transition from the ordered phases to Mixed Equilibrium



(a) Classic: $\partial L/\partial\gamma$.

(b) Mean-field: $\partial L/\partial\gamma$.

(c) Classic: $\partial L/\partial T$.

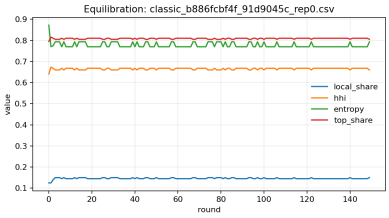
(d) Mean-field: $\partial L/\partial T$.

Figure 12. Susceptibility maps for steady-state local share L^*, computed via second-order central differences (first-order at grid boundaries) from the 21×21 sweep. Top row: $\chi_\gamma = \partial L^*/\partial\gamma$, whose ridges locate the critical bandwidth $\gamma^*(T)$ separating Digital from Local Dominance. Bottom row: $\chi_T = \partial L^*/\partial T$, whose ridges trace $T^*(\gamma)$ separating the ordered phases from Mixed Equilibrium. The classic model exhibits sign changes in χ_T across the grid, indicating competing mechanisms absent from the mean-field solution.

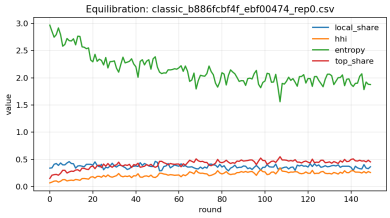
as T increases. The ridge is sharpest at low γ and broadens at high γ, where Local Dominance yields to Mixed Equilibrium more gradually. The critical temperature $T^*(\gamma)$ increases monotonically with γ: higher information access deepens the local-dominance potential well, requiring larger thermal fluctuations to destabilise it. The two surfaces together partition the (T, γ) plane into three sectors, providing phase-boundary estimates independent of the qualitative thresholds applied to L^* directly.

3.8 Mean-Field and Agent-Based Model Comparison

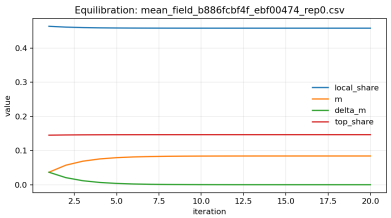
Figure 13 illustrates the qualitative difference between the two engines through representative trajectories: the ABM can either lock in quickly to a concentrated state (low T) or sustain persistent stochastic fluctuations around its stationary mean (higher T), while the mean-



(a) Classic: near-stationary trace (low noise).



(b) Classic: persistent stochastic fluctuations.



(c) Mean-field: smooth fixed-point convergence.

Figure 13. Representative simulation trajectories illustrating the qualitative difference between the two engines. At low noise (top left), the ABM locks in rapidly to a concentrated state and remains nearly flat. At higher noise (top right), vendor shares fluctuate visibly around their late-time mean even after the system is stationary—a finite-N effect absent from the mean-field solution. The mean-field engine (bottom) converges monotonically to its fixed point and does not exhibit temporal sampling noise.

field engine converges monotonically to its fixed point and exhibits no temporal sampling noise.

Figure 14 provides the quantitative comparison across the full grid. Within the bulk of both ordered phases and throughout the Mixed-Equilibrium region, the two engines agree to within the replicate-to-replicate variability of the ABM ($R = 10$ replicates), consistent with the law-of-large-numbers equivalence established in Section 2. Near the boundary separating Digital from Local Dominance at low T, however, the ABM exhibits path-dependent behaviour absent from the mean-field solution. Because the knowledge-accumulation process generates different consideration-set realisations across independent seeds, the ABM can converge to different long-run fixed points in the vicinity of $\gamma^*(T)$. This bistability is a finite-size signature of the first-order-like transition identified by the χ_γ ridge, and it is suppressed as $N \to \infty$. The mean-field engine reports the dominant fixed point alone; the bistable region is therefore identified more cleanly from ABM replicate

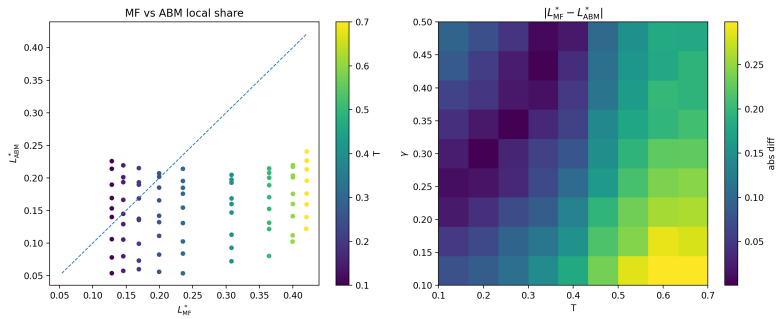


Figure 14. Quantitative comparison of mean-field and ABM engines across the full 441-point grid. *Left:* Scatter of L^*_{MF} against L^*_{ABM}, coloured by temperature T. Deviations are largest at intermediate T, near the phase boundary. *Right:* Heatmap of absolute discrepancy $|L^*_{\mathrm{MF}} - L^*_{\mathrm{ABM}}|$ in the (T, γ) plane. Maximal discrepancy concentrates along the susceptibility ridges of Figure 12, where slow mixing between metastable states prevents the ABM from converging to the mean-field fixed point within the simulation horizon.

variance than from the mean-field solution.

Pilot validation at fixed γ. A targeted preliminary sweep at $\gamma = 0.3$ over three temperature values ($T \in \{0.1, 0.5, 1.0\}$, $R = 5$ replicates) confirms the systematic nature of the ABM–mean-field divergence quantified above (Tables 1 and 2). At $\gamma = 0.3$ the market remains digital-dominant in the ABM across the full tested range: late local share rises only from 0.147 ± 0.011 at $T = 0.1$ to 0.248 ± 0.031 at $T = 1.0$. The mean-field engine predicts a far stronger thermalisation: late local share reaches 0.441 ± 0.000 at $T = 1.0$ and HHI collapses from 0.773 to 0.104 over the same range, while the ABM sustains HHI values of 0.485–0.633. Entropy and top-vendor share show the same pattern: mean-field rapidly approaches the high-diversity, low-concentration Mixed Equilibrium prediction, while the ABM sustains winner-take-most structure driven by reinforcement and finite-N stochasticity. Temperature susceptibility χ_T is largest at low T in mean-field (0.588 at $T = 0.1$) but roughly flat and much smaller in the ABM (0.118–0.132), consistent with a smooth crossover rather than a sharply localised critical ridge at this γ value.

3.9 Sensitivity to Social Coupling Strength

Figure 15 examines how the qualitative character of the phase transition changes as J varies over $[0.0, 0.8]$ at fixed $\gamma = 0.5$. At $J = 0$ the utility contains no social coupling and the T-driven transition is continuous with a broad susceptibility peak. As J increases, the fixed-

Table 1. Pilot sweep at $\gamma = 0.3$: late-window order parameters (means $\pm$ s.d., $R = 5$ replicates).

Model	T	L	HHI	Entropy	Top share
ABM	0.1	0.147 ± 0.011	0.633 ± 0.045	0.510 ± 0.075	0.794 ± 0.032
ABM	0.5	0.206 ± 0.049	0.607 ± 0.055	0.586 ± 0.090	0.754 ± 0.034
ABM	1.0	0.248 ± 0.031	0.485 ± 0.020	1.066 ± 0.193	0.674 ± 0.045
Mean-field	0.1	0.129 ± 0.000	0.773 ± 0.000	0.420 ± 0.000	0.867 ± 0.000
Mean-field	0.5	0.364 ± 0.000	0.313 ± 0.000	1.704 ± 0.000	0.484 ± 0.000
Mean-field	1.0	0.441 ± 0.000	0.104 ± 0.000	2.733 ± 0.000	0.241 ± 0.000

Table 2. Pilot sweep at $\gamma = 0.3$: temperature susceptibility $\chi_T = \partial L/\partial T$ by finite differences (means $\pm$ s.d., $R = 5$ replicates).

Model	T	χ_T
ABM	0.1	0.118 ± 0.071
ABM	0.5	0.127 ± 0.025
ABM	1.0	0.132 ± 0.072
Mean-field	0.1	0.588 ± 0.000
Mean-field	0.5	0.395 ± 0.000
Mean-field	1.0	0.154 ± 0.000

point map $m = \mathcal{F}(m; T, \gamma, J)$ of equation (3) becomes increasingly nonlinear in m and $T^*(J)$ decreases monotonically: the ordered phase is destabilised at lower temperatures by the growing cost of resisting the social field. Above a critical coupling $J_c(\gamma = 0.5)$, two stable fixed points coexist at intermediate T and the mean-field solution jumps discontinuously as T crosses T^*, signalling a first-order-like transition.

This behaviour has a direct economic interpretation. Weak social coupling ($J < J_c$) corresponds to a market in which vendor popularity has a modest but non-decisive influence on choice; such a market transitions smoothly between phases as external conditions change. Strong social coupling ($J > J_c$) corresponds to a market in which popularity constitutes self-reinforcing evidence of quality, producing genuine tipping dynamics: a small perturbation near the phase boundary can drive a rapid, large-amplitude shift in market structure that is difficult to reverse. The primary-sweep value $J = 0.2$ sits below J_c at $\gamma = 0.5$, ensuring that the phase boundaries reported in Section 3.7 reflect the geometry of the utility landscape rather than a strong discontinuity.

3.10 Summary of Market Phases

Table 3 consolidates the three identified market phases in terms of all four order parameters. The critical loci $T^*(\gamma)$ and $\gamma^*(T)$ derived from the susceptibility analysis, together with the order-parameter maps of

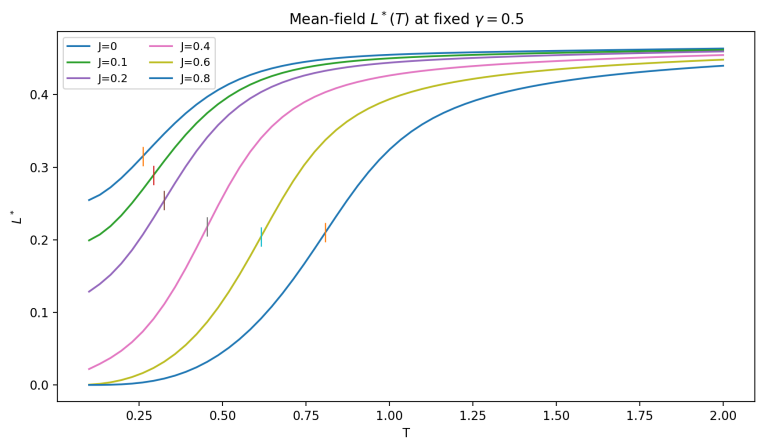


Figure 15. Effect of social coupling strength J on the T-dependence of equilibrium local share L^* at fixed $\gamma = 0.5$, from the mean-field engine. At $J = 0$ the transition is continuous and broad. As J increases the transition sharpens; above a critical coupling $J_c(\gamma = 0.5)$ the transition becomes discontinuous, exhibiting a jump ΔL^* at $T^*(J)$. Tick marks indicate T^* per curve; T^* decreases monotonically with J.

Phase	m^*	L^*	H^*	HHI*
Local Dominance	< -0.2	> 0.6	Low	Intermediate
Digital Dominance	$> +0.2$	< 0.4	Low	High
Mixed Equilibrium	≈ 0	≈ 0.5	High	Low

Table 3. Summary of the three market phases identified in the (T, γ) phase diagram. Threshold values for m^* and L^* are operational definitions anchored to the $|\chi| = 0.75 \max |\chi|$ susceptibility-ridge boundaries of Figure 12. The qualitative descriptors for H^* and HHI* are relative to the theoretical bounds $[0, \log M]$ and $[1/M, 1]$ for $M = 40$.

Figures 7–11, constitute the central quantitative results of this study.

4. Discussion

This study frames platform–local competition as a thermodynamic control problem in which macroscopic market structure is organised by a small set of global parameters. Across both engines, the phase diagram in the (T, γ) plane exhibits the same qualitative decomposition into (i) Digital Dominance at low information access, (ii) Local Dominance when local discovery is sufficiently strong, and (iii) a high-noise Mixed Equilibrium in which utility gradients are thermally suppressed. The key interpretive point is that *information access and decision noise*

play structurally distinct roles: γ controls whether local vendors enter the consideration set at all, while T controls the sharpness of choice among the currently visible options. This separation is visible in the order-parameter maps (Figures 7 and 8) and is reinforced by the secondary metrics: as γ increases, the ABM exhibits a robust decrease in market concentration (HHI) and a corresponding increase in entropy even in regions where L^* changes gradually (Figure 9). In practical terms, information access diversifies demand by broadening the set of competitors, whereas temperature diversifies demand by smoothing the relative advantage among those competitors.

A central finding is that the ABM and mean-field engine do not simply differ by finite-size noise; they differ in *which control parameter is dynamically active* in the boundary region. In the ABM, γ is an active dynamical control because the accumulating-knowledge mechanism converts per-round discovery into a path-dependent expansion of the consideration set. This manifests as a strong and spatially structured $\chi_\gamma = \partial L^*/\partial\gamma$ surface, with localised hotspots rather than a single smooth ridge (Figure 10). Such hotspots are consistent with a threshold mechanism: once a sufficient fraction of local vendors has been discovered by enough customers, marginal increases in information access produce disproportionately large gains in local adoption. In contrast, the mean-field engine freezes the knowledge sets prior to solving the fixed point. As a result, γ is largely reduced to a static parameter controlling a single precomputed mask, and the resulting χ_γ surface is near zero across most of the zoomed plane (Figure 10). The mean-field model therefore captures equilibrium mixing under a given visibility state, but it does not faithfully reproduce the ABM's information-driven transition mechanism in regimes where discovery and reinforcement interact over time. This is not a minor numerical discrepancy; it is a modelling discrepancy induced by the static-mask approximation.

The temperature axis provides a second, complementary contrast. In both engines, increasing T reduces concentration and increases entropy, consistent with the Boltzmann interpretation of discrete choice. However, the ABM sustains substantially more winner-take-most structure at moderate and high temperature than the mean-field prediction, as illustrated by the pilot $\gamma = 0.3$ slice (Section 3.5) and by the qualitative trajectories (Figure 13). The ABM can remain concentrated because the utility includes reinforcement and a bandwagon term, both of which create persistent effective fields that are not averaged away by thermal noise over a finite horizon. The mean-field engine, by omitting explicit affinity dynamics and by collapsing temporal sampling into an equilibrium computation, thermalises much more rapidly. From a statistical-mechanical perspective, the ABM is sampling a rugged and evolving effective energy landscape (memory and popularity terms

change the landscape over time), while the mean-field engine solves a smoothed stationary landscape. The consequence is that mean-field can correctly predict monotonic trends with T while systematically underestimating concentration and overestimating diversity in regimes where the ABM exhibits slow mixing and metastability.

The susceptibility analysis provides a compact way to interpret these behaviours as transition-like responses rather than isolated numerical outcomes. In the coarse 21×21 sweep, ridges in χ_γ and χ_T partition the plane into three sectors (Figure 12), consistent with the interpretation of phase boundaries as loci of maximal responsiveness of the order parameter. In the zoomed survey, the mean-field χ_T map exhibits a coherent band of high response at intermediate T (Figure 10), consistent with an Ising-like crossover where the system transitions from an ordered state to a mixed state as thermal disorder increases. In the ABM, χ_T is noisier and exhibits sign changes and patchiness (Figure 12), which is expected when finite differences are taken on late-window averages in a stochastic model, but also suggests competing mechanisms: temperature simultaneously increases exploration (which tends to diversify shares) and can weaken reinforcement-driven lock-in at different rates depending on the realised discovery history. The existence of a well-defined mean-field ridge but a patchier ABM response indicates that the ABM's boundary region contains a mixture of slow-mixing trajectories rather than a single sharp critical manifold at this coupling strength.

The coupling-sensitivity experiment clarifies when true tipping behaviour should be expected. As J increases (Figure 15), the mean-field transition sharpens and can become discontinuous, consistent with the emergence of multiple stable fixed points in the Curie–Weiss mapping. Economically, this corresponds to a market where popularity or type-level social influence becomes decisive: once an advantage emerges, it is amplified by self-consistent feedback. The primary sweep at $J = 0.2$ sits in a regime where transitions are predominantly continuous and broadened by stochasticity, which is consistent with the smooth gradients observed in the ABM local-share surface (Figures 7 and 8). At stronger coupling, the same framework predicts genuine hysteresis and bistability, making the phase boundary operationally important for policy or design because small perturbations can trigger large irreversible shifts.

Finally, the results provide a policy-relevant interpretation of "market stability" in terms of the phase geometry. Digital Dominance at low γ is not merely the outcome of better digital utility; it is structurally enforced by visibility: if local vendors cannot enter the consideration set, thermal noise cannot rescue them. Mixed Equilibrium at high T is stable in the opposite sense: diversity is maintained even when utilities differ, because decision noise prevents persistent concentration.

Between these extremes, the system resides in a sensitive region where γ controls whether local vendors can accumulate sufficient visibility to compete and where T controls whether small utility differences become amplified into winner-take-most outcomes. The dual-engine approach therefore yields a physically interpretable diagnostic: mean-field locates the equilibrium geometry of the utility landscape, while the ABM reveals where real dynamics—discovery, memory, and finite-population fluctuations—shift, broaden, or fragment the predicted phase boundaries.

5. Limitations

Several limitations of the present framework warrant explicit acknowledgement.

Finite-size effects. The ABM is run at $N = 200$ customers and $M = 40$ vendors. At this scale, deviations from the mean-field limit scale as $\mathcal{O}(N^{-1/2})$ in the bulk of each phase and are amplified near phase boundaries where susceptibility is large. Systematic finite-size scaling studies—repeating the phase sweep at increasing population sizes and extrapolating order parameters and susceptibility peaks to $N \to \infty$—would sharpen the location of phase boundaries and confirm whether the first-order-like features observed at the χ_γ ridge survive in the thermodynamic limit.

Static-mask approximation in the mean-field engine. The mean-field fixed-point engine freezes each customer's consideration set prior to solving the self-consistency equation, replacing the round-by-round accumulating-knowledge dynamics with a single precomputed mask. This approximation is responsible for the near-zero χ_γ surface in the mean-field results and means the engine cannot reproduce the information-threshold dynamics that the ABM exhibits. A dynamic mean-field formulation—iterating both the consideration-set expansion and the fixed-point equation simultaneously—would close this gap but at substantially increased computational cost.

Fixed population structure and local fraction. All sweeps are performed at a fixed local fraction $\lambda = 0.5$ and a fixed total vendor count $M = 40$. The phase diagram is expected to shift quantitatively as λ changes—in particular, the $\gamma^*(T)$ boundary should move toward lower γ as local vendors become more numerous relative to digital vendors. The restriction to a symmetric bipartite structure with exogenously determined vendor attributes limits the model's applicability to markets where vendor pricing and positioning are determined outside the simulation horizon.

Absence of strategic vendor behaviour. Vendors in the present model are passive: they do not observe market shares, adjust prices, or invest in visibility. Real digital platforms engage in algorithmic ranking,

targeted promotion, and dynamic pricing that can shift the effective γ and T experienced by different customer segments. Incorporating vendor-level optimisation as a Stackelberg or Nash equilibrium problem layered on top of the consumer choice model represents a natural but non-trivial extension.

Two-dimensional unit-square geography. The spatial structure is a stylised unit square with uniformly distributed local vendors and Gaussian-distributed customers. Real urban geographies exhibit clustering, street-network constraints, and heterogeneous vendor density. The calibrated logistic visibility kernel mitigates some of this limitation by anchoring discovery probability to empirical percentiles of the distance distribution, but the underlying topology is nonetheless a significant abstraction.

Fixed-population cache across sweeps. All parameter sweeps reuse the same realised customer and vendor population, isolating control-parameter variation from population-level randomness. This is appropriate for clean phase-diagram construction, but it means the reported phase boundaries are conditional on a particular market realisation. Population-averaged results—sweeping over multiple independent population draws in addition to control parameters—would quantify the sensitivity of the phase diagram to the underlying market structure.

6. Conclusion

This paper has developed and analysed a statistical-mechanics-inspired model of consumer choice between local and digital vendors, using a dual-engine simulation framework to map macroscopic market structure across the (T, γ) control plane. Three qualitatively distinct phases were identified and characterised through order parameters, entropy, and susceptibility: Digital Dominance at low information bandwidth, Local Dominance at high information bandwidth, and a Mixed Equilibrium at high decision noise. Phase boundaries were located quantitatively via peaks in $\chi_\gamma = \partial L^*/\partial\gamma$ and $\chi_T = \partial L^*/\partial T$, establishing that the (T, γ) plane admits a coherent thermodynamic interpretation in which the two control parameters play structurally distinct and complementary roles.

The primary methodological contribution is the demonstration that the ABM and mean-field engine differ not merely in noise level but in which control parameter is dynamically active. The accumulating-knowledge mechanism in the ABM causes γ to act as a genuine path-dependent dynamical control, producing spatially localised χ_γ hotspots that signal an information-access threshold: once a sufficient fraction of local vendors has been discovered, marginal improvements in information access translate into disproportionately large gains in local adoption. The mean-field static-mask approximation suppresses this

mechanism, rendering χ_γ near zero across the zoomed plane and systematically overpredicting market thermalisation at moderate and high T. This is a modelling discrepancy, not a finite-size artefact, and it identifies the boundary of validity of equilibrium mean-field analysis for information-driven market transitions. The coupling-sensitivity experiment further establishes that genuine tipping dynamics—with hysteresis and discontinuous market shifts—emerge when the Ising-type social coupling J exceeds a critical value $J_c(\gamma)$, providing a precise criterion for when platform competition becomes self-reinforcing.

These results carry interpretable policy implications. Digital Dominance at low γ is structurally enforced by visibility rather than by superior utility: local vendors that cannot enter the consideration set are excluded regardless of their price or quality advantages. Interventions that raise γ—open-protocol discovery infrastructure, standardised search APIs, or neutral aggregator listing requirements—function as effective shifts of the phase boundary, potentially tipping a Digital-Dominance market into a Local-Dominance or Mixed regime without requiring any change in the underlying utility structure. The threshold character of χ_γ means such interventions have strongly nonlinear effects: below the threshold, marginal increases in information access produce little change; above it, the same marginal increase can trigger a rapid structural transition.

Future work should address the limitations enumerated in Section 5, particularly finite-size scaling to sharpen the thermodynamic-limit phase diagram, a dynamic mean-field formulation that captures the accumulating-knowledge mechanism, and the incorporation of endogenous vendor strategy. The statistical-mechanics framework developed here provides a natural language for all of these extensions, and the dual-engine design provides a ready methodology for validating analytic approximations against explicit agent dynamics.

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